# 1. Introduction

A lithium-ion battery is a rechargeable electrochemical energy storage device. During discharge, lithium ions move from the negative electrode (anode) through an electrolyte to the positive electrode (cathode); charging reverses the flow. The technology was commercialised by Sony in 1991, building on the work of John Goodenough, Stanley Whittingham and Akira Yoshino, who shared the 2019 Nobel Prize in Chemistry. Today lithium-ion cells power phones, laptops, electric vehicles and grid storage systems.

# 2. Cell components

Every cell contains four essential components. The anode is usually graphite. The cathode is a lithium metal oxide or phosphate. The electrolyte is a lithium salt, typically LiPF6, dissolved in organic carbonate solvents. A porous polymer separator keeps the electrodes apart to prevent short circuits while still letting lithium ions pass through.

# 3. Cathode chemistries

Common cathode families are lithium cobalt oxide (LCO), nickel manganese cobalt oxide (NMC), nickel cobalt aluminium oxide (NCA), lithium iron phosphate (LFP) and lithium manganese oxide (LMO). NMC cells offer high energy density of roughly 200-250 Wh/kg at cell level. LFP cells have a lower energy density of about 160 Wh/kg, but they last longer, are more thermally stable and contain no cobalt.

## 4. Performance metrics

The key metrics are energy density (Wh/kg), specific power, cycle life and C-rate. Cycle life is the number of full charge-discharge cycles a cell completes before its capacity falls to 80% of the original value. NMC cells typically deliver 1,000-2,000 cycles, while LFP cells reach 2,000-4,000 cycles. The nominal voltage is about 3.6-3.7 V for NMC cells and 3.2 V for LFP cells.

## 5. Safety and thermal runaway

Lithium-ion cells can fail through thermal runaway, a self-accelerating chain of exothermic reactions that begins when the internal temperature rises beyond roughly 150 °C. A battery management system (BMS) monitors voltage and temperature of every cell group and disconnects the pack if limits are exceeded. Separators with a shutdown layer add a second line of defence.

## 6. Cost trends

Battery pack prices have fallen by roughly 90% since 2010. BloombergNEF reported an average pack price of $139 per kWh in 2023, a level that brings electric vehicles close to price parity with petrol cars in several markets.

## 7. Outlook

Solid-state batteries replace the liquid electrolyte with a solid one and may reach energy densities of up to 400 Wh/kg. Their commercial introduction is expected by 2027. Sodium-ion cells are being developed as a lower-cost complement for stationary storage, since sodium is far more abundant than lithium.